

LAB EXPERIMENT RECORD

Set 5: Website Analytics

Department	B.Tech AIDS
Course Code / Course Name	DSA06 / Data Handling and Visualization
Experiment	Data Visualization using R
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Aim

To analyze website traffic data using a line chart, bar chart, and stacked area chart in R, and to build an interactive dashboard combining the line chart and bar chart using R Shiny.

Dataset: Website Traffic

Date	Page Views	Click-through Rate
2023-01-01	1500	2.3%
2023-01-02	1600	2.7%
2023-01-03	1400	2.0%
2023-01-04	1650	2.4%
2023-01-05	1800	2.6%

Note: The experiment table does not contain user interaction values. For Question 3, the R program uses the following values for 01 to 05 January: Likes – 120, 150, 100, 130, 170; Shares – 45, 60, 30, 50, 65; Comments – 25, 35, 15, 20, 40.

PROGRAM 1: Line Chart of Daily Page Views

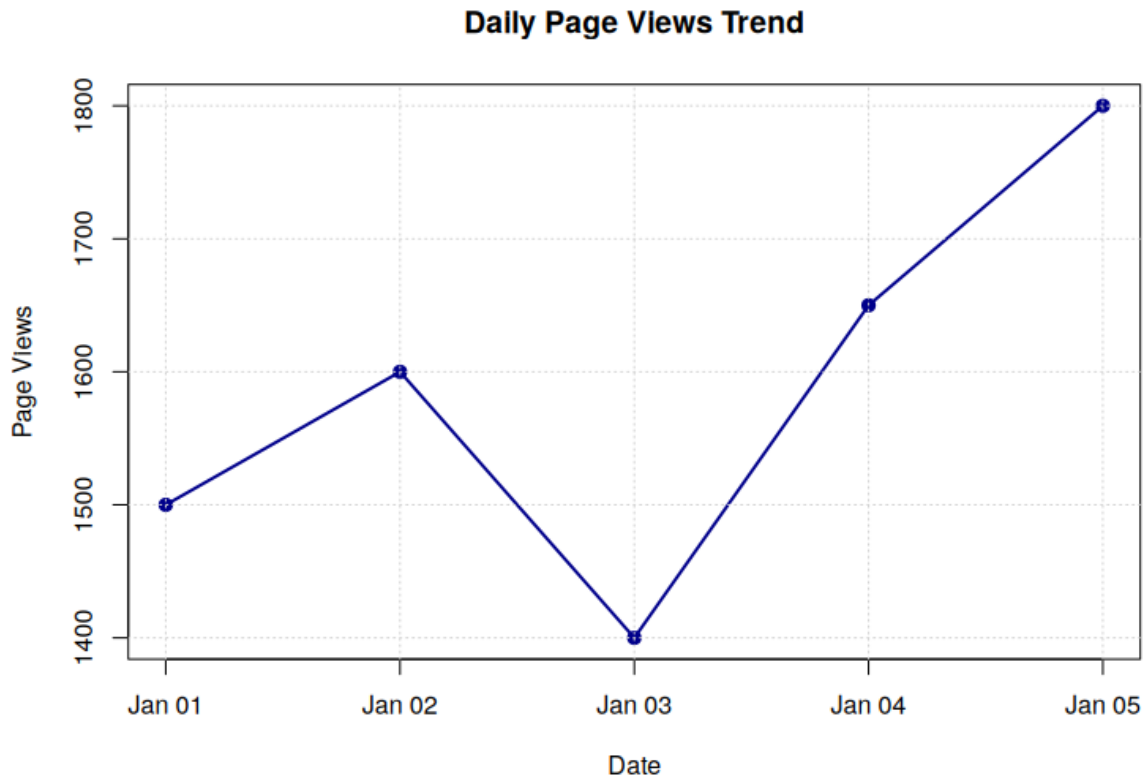
Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

```
# Create dataset for Set 5
website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6),
  Likes = c(120, 150, 100, 130, 170),
  Shares = c(45, 60, 30, 50, 65),
  Comments = c(25, 35, 15, 20, 40)
)

print(website_data)

# 1. Line Chart for Page Views Trend
```

```
plot(website_data$Date, website_data$Page_Views,  
     type = "o",  
     col = "darkblue",  
     lwd = 2,  
     pch = 19,  
     xlab = "Date",  
     ylab = "Page Views",  
     main = "Daily Page Views Trend")  
  
grid()
```



Output 1: Daily Page Views Trend

Observation: Page views rise from 1500 to 1600, drop to the lowest value of 1400 on 03 January, and then increase steadily to the highest value of 1800 on 05 January.

PROGRAM 2: Bar Chart of Top Days by CTR

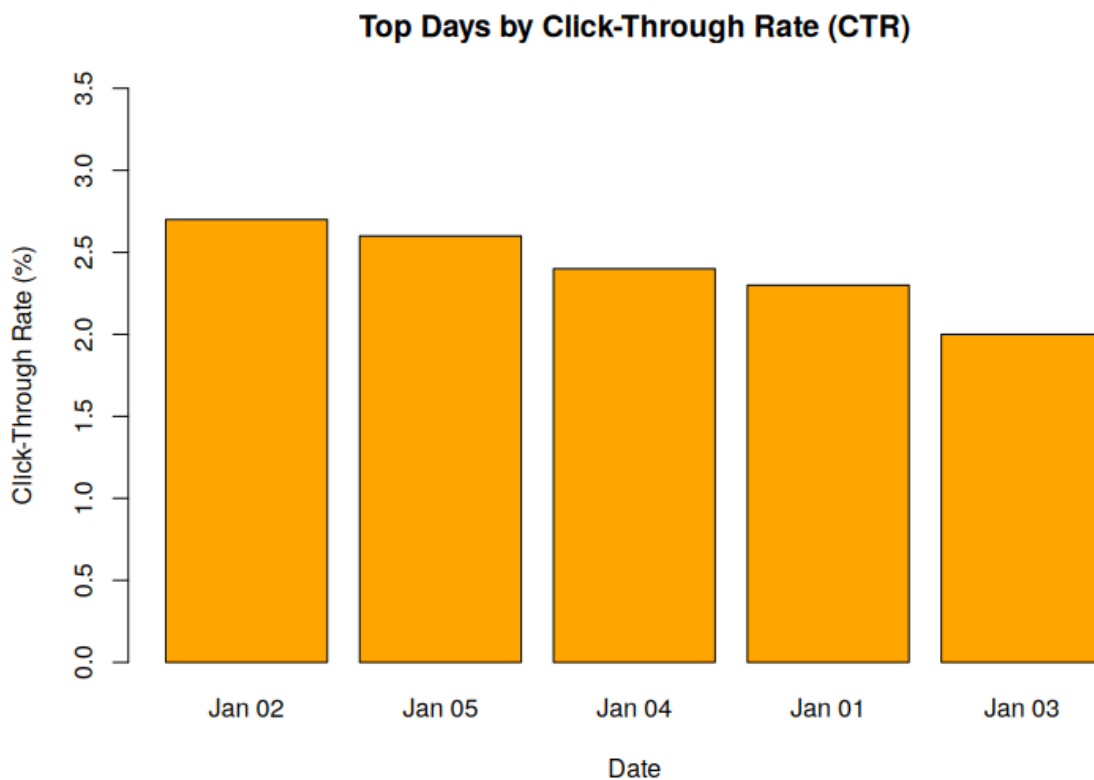
Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

```
# Create dataset for Set 5
website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6),
  Likes = c(120, 150, 100, 130, 170),
  Shares = c(45, 60, 30, 50, 65),
  Comments = c(25, 35, 15, 20, 40)
)

print(website_data)

# Sort data by CTR in descending order
sorted_ctr <- website_data[order(-website_data$CTR), ]

# 2. Bar Chart for Highest CTR Days
barplot(sorted_ctr$CTR,
  names.arg = format(sorted_ctr$Date, "%b %d"),
  col = "orange",
  xlab = "Date",
  ylab = "Click-Through Rate (%)",
  main = "Top Days by Click-Through Rate (CTR)",
  ylim = c(0, 3.5))
```



Output 2: Top Days by Click-Through Rate

Observation: The days are arranged from highest to lowest CTR. 02 January has the highest click-through rate (2.7%), followed by 05 January (2.6%) and 04 January (2.4%). 03 January has the lowest CTR (2.0%), which is also the day with the lowest page views.

PROGRAM 3: Stacked Area Chart of User Interactions

Develop a stacked area chart to display the distribution of user interactions (likes, shares, comments) on a website.

```
# Create dataset for Set 5
website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6),
  Likes = c(120, 150, 100, 130, 170),
  Shares = c(45, 60, 30, 50, 65),
  Comments = c(25, 35, 15, 20, 40)
)

print(website_data)

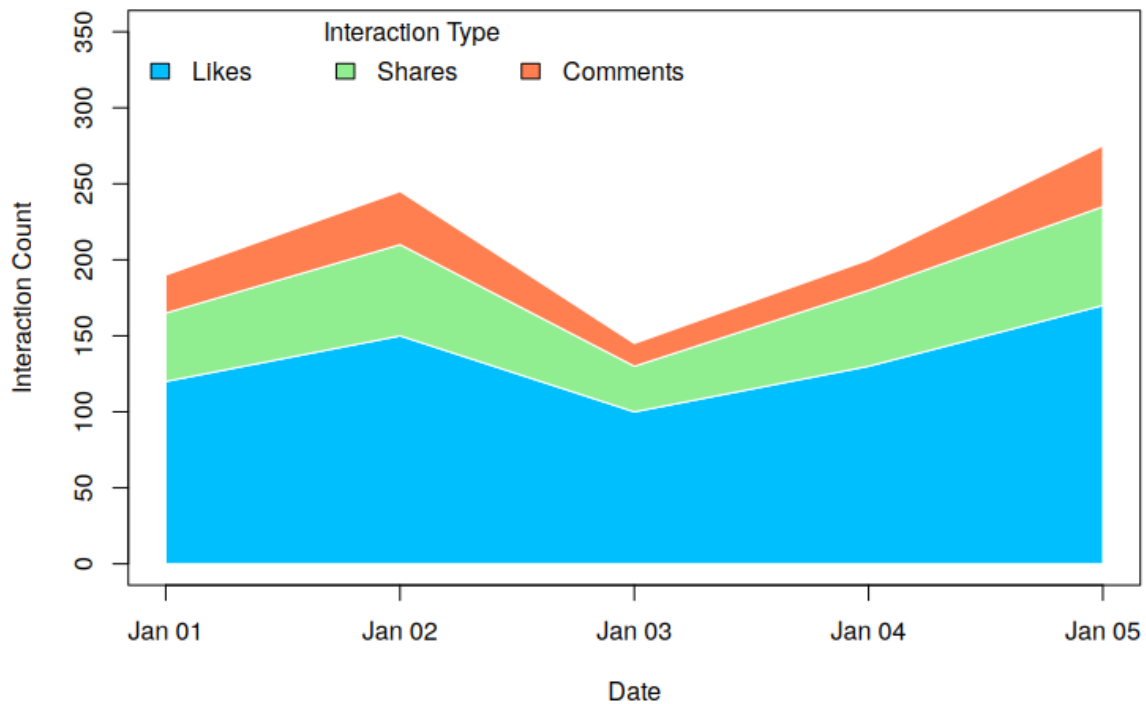
# Cumulative values for stacking (Likes, Shares, Comments)
day <- 1:nrow(website_data)
lay1 <- website_data$Likes
lay2 <- lay1 + website_data$Shares
lay3 <- lay2 + website_data$Comments
area_cols <- c("deepskyblue", "lightgreen", "coral")

# 3. Stacked Area Chart
plot(day, lay3, type = "n",
     xaxt = "n",
     ylim = c(0, 350),
     xlab = "Date",
     ylab = "Interaction Count",
     main = "Distribution of User Interactions Over Time")
axis(1, at = day, labels = format(website_data$Date, "%b %d"))

polygon(c(day, rev(day)), c(lay1, rep(0, length(day))), col = area_cols[1], border = "white")
polygon(c(day, rev(day)), c(lay2, rev(lay1)), col = area_cols[2], border = "white")
polygon(c(day, rev(day)), c(lay3, rev(lay2)), col = area_cols[3], border = "white")

legend("topleft",
     legend = c("Likes", "Shares", "Comments"),
     fill = area_cols,
     title = "Interaction Type",
     horiz = TRUE, bty = "n")
```

Distribution of User Interactions Over Time



Output 3: Distribution of User Interactions Over Time

Observation: Likes form the largest part of user interactions on every day, followed by shares and then comments. Total interactions are highest on 05 January (275) and lowest on 03 January (145), following the same pattern as page views.

PROGRAM 4: Interactive Dashboard for Website Traffic

Build a Tableau dashboard with the line chart and bar chart for interactive exploration of website traffic data.

Note: Tableau could not be installed on the system used for this experiment, so the interactive dashboard was developed in R using the Shiny package, with the same charts and interactive filters required in the question.

```
library(shiny)

# Dataset for Set 5
website_data <- data.frame(
  Date = as.Date(c("2023-01-01", "2023-01-02", "2023-01-03", "2023-01-04", "2023-01-05")),
  Page_Views = c(1500, 1600, 1400, 1650, 1800),
  CTR = c(2.3, 2.7, 2.0, 2.4, 2.6)
)
day_labels <- format(website_data$Date, "%b %d")

ui <- fluidPage(
  titlePanel("Website Traffic Dashboard"),
  sidebarLayout(
    sidebarPanel(width = 3,
      checkboxGroupInput("days", "Dates:", choices = day_labels, selected = day_labels),
      selectInput("topn", "Top N days by CTR:", choices = 1:5, selected = 3)
    ),
    mainPanel(width = 9,
      fluidRow(
        column(6, plotOutput("linePlot", height = "340px")),
        column(6, plotOutput("barPlot", height = "340px"))
      )
    )
  )
)

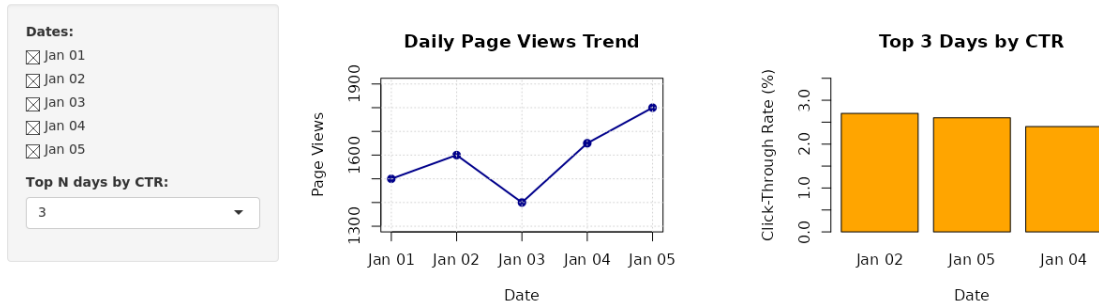
server <- function(input, output) {
  filtered <- reactive({ website_data[day_labels %in% input$days, ] })

  output$linePlot <- renderPlot({
    d <- filtered()
    plot(d$Date, d$Page_Views, type = "o", col = "darkblue", lwd = 2, pch = 19,
      ylim = c(1300, 1900), xlab = "Date", ylab = "Page Views",
      main = "Daily Page Views Trend")
    grid()
  }, width = 420, height = 340, res = 96)

  output$barPlot <- renderPlot({
    d <- filtered()
    top <- head(d[order(-d$CTR), ], as.numeric(input$topn))
    barplot(top$CTR, names.arg = format(top$Date, "%b %d"), col = "orange",
      ylim = c(0, 3.5), xlab = "Date", ylab = "Click-Through Rate (%)",
      main = paste("Top", nrow(top), "Days by CTR"))
  }, width = 420, height = 340, res = 96)
}

shinyApp(ui = ui, server = server)
```

Website Traffic Dashboard



Output 4: Website Traffic Interactive Dashboard (R Shiny)

Observation: The dashboard shows the daily page-view trend and the top N days by click-through rate side by side. The Dates filter selects which days are shown, and the Top N box changes how many days appear in the bar chart. With N = 3, the top days are 02 January (2.7%), 05 January (2.6%), and 04 January (2.4%).

Procedure

1. Create the website traffic data frame with Date, Page Views, CTR, Likes, Shares, and Comments.
2. Use `plot()` with `type = "o"` to draw the page-view line chart and `grid()` to add grid lines.
3. Sort the data by CTR using `order()` and draw the bar chart using `barplot()`.
4. Calculate cumulative values of likes, shares, and comments and draw the stacked areas using `polygon()`.
5. Install and load the shiny package, and create the dashboard layout using `fluidPage()`, sidebar filters, and output areas for the charts.
6. Write the server function to filter the data with `reactive()` and redraw the outputs, then run the dashboard using `shinyApp()` and test the filters.

Result

The website traffic data was successfully visualized using a line chart, bar chart, and stacked area chart in R, and an interactive R Shiny dashboard was built combining the line chart and bar chart.

Key R Functions Used

Function	Purpose
<code>as.Date()</code>	Converts text values into dates.
<code>plot()</code>	Creates the line chart and the base for the area chart.
<code>order()</code>	Sorts the days by click-through rate.
<code>barplot()</code>	Creates the CTR bar chart.
<code>polygon()</code>	Draws the filled layers of the stacked area chart.
<code>legend()</code>	Adds a legend for the interaction types.
<code>fluidPage()</code> / <code>checkboxGroupInput()</code>	Creates the dashboard layout and filters.
<code>reactive()</code> / <code>renderPlot()</code>	Filters the data and redraws the charts.
<code>shinyApp()</code>	Runs the interactive dashboard.